

Théo Riou

Curriculum Vitae

72 Avenue de Lardenne
31100 Toulouse, France
+33 (0) 6 73 33 69 97
theo.riou@imft.fr

Personal Information

Date of Birth June 18, 2000 in France
Personal Page <https://thriou.github.io/>

Education

- 11/2023– **Ph.D in Combustion**, *IMFT, Université de Toulouse*, France.
Topic: Numerical simulation of transient dynamics in an aircraft injector for hydrogen combustion: Anchoring and lift-off dynamics, longitudinal and transverse thermoacoustic instabilities, combustion noise.
Supervisors: Laurent Selle, Guillaume Daviller.
- 2022–2023 **Master 2 Combustion**, *ENSMA*, Poitiers, France.
◦ *Turbulent combustion, Combustion in twphase flows*
- 2020–2023 **Engineering degree**, *ENSMA*, Poitiers.
◦ *Combustion, numerical scheme, probability, turbulence, CFD for industry etc. MET-TRE LES COURS DE NUMERIAUE ET DE MATH*
- 2018–2020 **Classe préparatoire aux grandes écoles MPSI/MP**, *Victor Grignard*, France.
- 2018 **Baccalauréat scientifique**, *Victor Grignard*, France.
◦ *Mention Très Bien*

Experience

- 04/2023 – 09/2023 **Internship**, *CEA, Paris Saclay*, France.
◦ *Simulation of a centrifugal pump using Neptune CFD to validate two-phase flow conditions.*
- 06/2022 – 08/2022 **Internship**, *University of Leeds*, England.
◦ *Calibration of a turbulence model for viscoelastic fluids to reduce drag in pipes using OpenFoam.*

Scientific contributions

- [1] Riou *et al.* Numerical simulation of flame lift-off dynamics in a hydrogen dual-swirl burner, *Proc. Combust. Inst.*, 2026 (accepted)
- [2] McDermott *et al.* An open-source anisotropic $k-\epsilon-v^2-f$ model for turbulent viscoelastic duct flows, *Phys. Fluids* (2023), doi: 10.1063/5.0159668

Talks and Poster sessions

- 07/27–31/2026 **Poster**, *ISOC*, Kyoto, Japan.
- 11/27–28/2025 **Talk**, *JTCH*, Saclay, France.
- 06/05/2025 **Poster**, *JDD*, Toulouse, France.
- 04/07–11/2025 **Poster**, *ECM*, Edinburgh, Scotland.
- 02/26–29/2024 **Talk**, *Hydrogen week*, Toulouse, France.

Grants and Honors

- 2025 **HPC allocation**, *PRACE*
◦ *Together with two colleagues, secured a European HPC computing grant based on a proposal (70 millions CPU hours).*

Responsibilities

- 2025-2026 **Weekly seminars organisation**, IMFT.
2026 **International seminar organisation**, *Hylon Days*, IMFT.
2024 **International seminar organisation**, *Hydrogen week: towards high pressure experiments and simulations*, IMFT.

Teaching

- 2026 **Internship supervision (M1)**, *ENSEEIH*T, Toulouse, France.
◦ *Internship on spectral analysis and modal decomposition for studying thermoacoustic instabilities in a hydrogen burner.*
2024-2025 **Teaching**, *ENSEEIH*T, Toulouse, France.
◦ *M1 turbulence practical works: hot wire, Particle Image Velocimetry*
◦ *M1 project: modal decomposition in order to study combustion noise*

Skills

- Language** French (native), English (B2, TOEIC : 890), Spanish (A2), German (A1).
Computer ◦ **Simulation software:** AVBP, Neptune CFD, Cantera, OpenFoam
◦ **Meshing:** Centaur, Salome
◦ **CAD:** Catia
◦ **Postprocessing:** Paraview
◦ **Other:** Python, Antares, Fortran, Linux, GeoGebra

Area Of Interest

- Music** piano (>19 years), national piano competitions, musical theory
Maths amateur research, reading
Philosophy/theology reading, formations (collège des Bernardins)
Sport self-guided hiking (mountains, GR etc.)

REFERENCES

- Laurent Selle e-mail: laurent.selle@imft.fr;